

CSE655 - Network Science - Assignment 1 - Report

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Question 1

part(a)

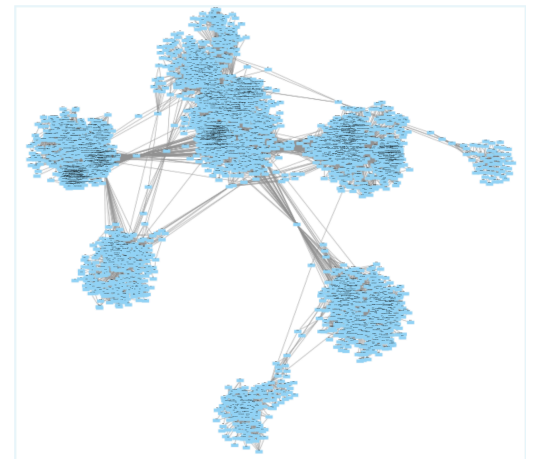
- **Methodology:** Read the dataset as an edge list (each line: u v). Stored edges as a list of tuples $\rightarrow (u, v)$. Constructed an adjacency matrix of size $N \times N$.
- **Results:**
 - Nodes (N): ~4039 and Edges (E): ~88,234. Adjacency matrix constructed successfully.

[illegible]

- **Observations:** The matrix is extremely large (4039×4039). The majority of entries are 0.
- **Interpretation:** The network is sparse; the adjacency matrix is memory-inefficient for large real networks. Edge lists/adjacency lists are better representations for sparse graphs.

part(b)

- **Methodology:** Created a CSV of the graph dataset. Used Cytoscape to plot the data. A force-directed layout was applied to reveal structural patterns.
- **Results:** The graph is shown on the side. It has:
 - Several dense clusters; a Highly connected central region. Bridge-like connections between clusters
 - Some nodes with visibly higher connectivity (hubs)
- **Observations:**
 - Community structure is visible. The network is not random; nodes form tightly connected groups.
 - A few nodes appear central and connect multiple clusters.
- **Interpretation:**
 - Community structure: typical in social networks.
 - Presence of hubs: consistent with heavy-tailed degree distribution. Small-world connectivity: clusters connected



part(c)

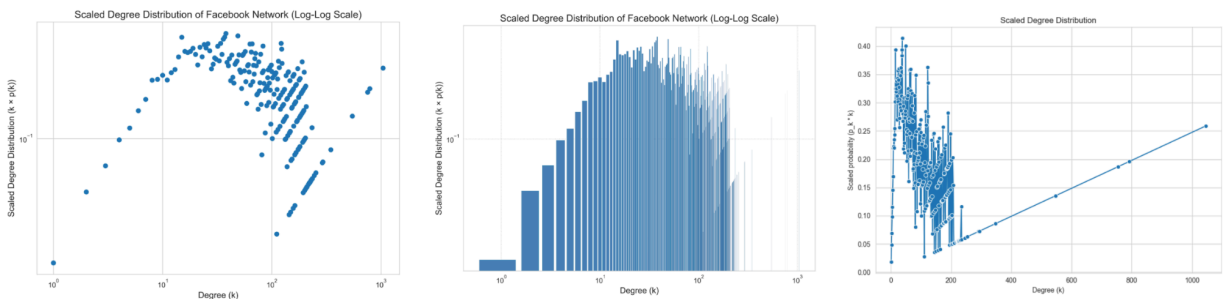
- **Methodology:** Computed Network density (Edges divided by max no. of Edges): $2E/N*(N-1)$
- **Results:** Network Density: 0.010819963503439287
- **Observations:** Only ~1% of all possible edges exist.
- **Interpretation:** The network is sparse. In real-world networks, the number of edges grows approximately linearly with N ($E \sim N$), not N^2 . This aligns with Barabási's discussion of real networks.

part(d)

- **Methodology:** For an undirected network: $\langle k \rangle = 2 \cdot E / N$ where: N = number of nodes E = number of edges
- **Results:** Average Degree $\langle k \rangle$: 43.69101262688784
- **Observations:** Each node connects to ~44 others on average.
- **Interpretation:** Despite being sparse globally, nodes still maintain moderate local connectivity, typical of social networks.

part(e)

- **Methodology:** Computed the degree of each node. Computed p_k (scaled degree distribution) and plotted the graphs below
- **Results:**



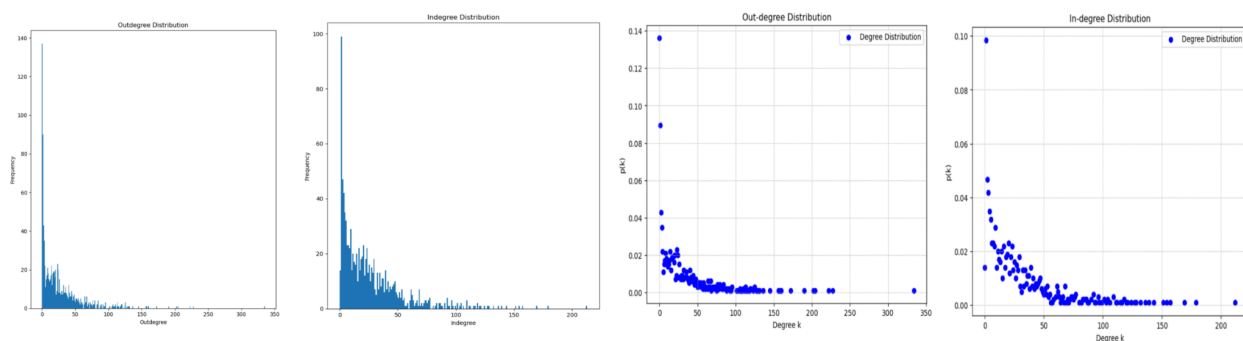
- **Observations:**
 - Distribution is highly skewed. Few nodes have a very large degree.
 - Many nodes have a small degree. Heavy-tailed behaviour is visible.
- **Interpretation:** The network exhibits scale-free characteristics, including hubs and Degree heterogeneity.

part(f)

- **Methodology:** Ran BFS from every node. Computed all shortest path distances. Averaged over all pairs.
- **Results:** Average Path Length: 3.6925, Diameter: 8, Average Clustering Coefficient: 0.6055
- **Observations:** Distances are surprisingly small.
- **Interpretation:** Small-world property: Even in large networks, nodes are only a few steps apart.

Question 2

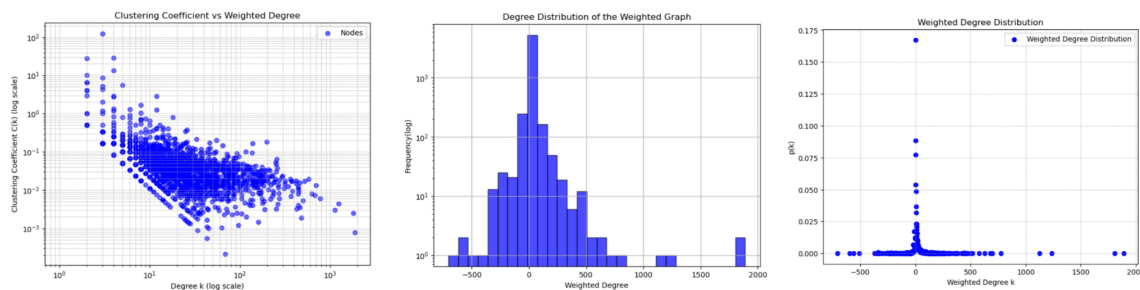
- **Methodology:** The dataset was modelled as a directed graph from an edge list. Arrays were used to compute indegree and outdegree by incrementing counts for each edge (u, v) . Histograms were then plotted to visualise the degree distributions.
- **Results and Observations:** Most nodes have low indegree and outdegree, while a few have much higher values. The distributions are skewed, showing uneven connectivity.



- **Interpretation:** The presence of nodes with high indegree or outdegree indicates role differentiation within the network, with certain nodes acting as more central or active participants. This behaviour is consistent with many real-world directed networks.

Question 3

- **Methodology:** The Bitcoin OTC network was modelled as a weighted directed graph. Weighted in-degree, out-degree, and total weighted degree were computed using dictionaries. The weighted degree distribution was normalised to obtain $P(k)$ and plotted on log scales. The weighted clustering coefficient was calculated and normalised. Finally, the clustering coefficient was plotted against weighted degree on a log–log scale to analyse network structure.
- **Results:**



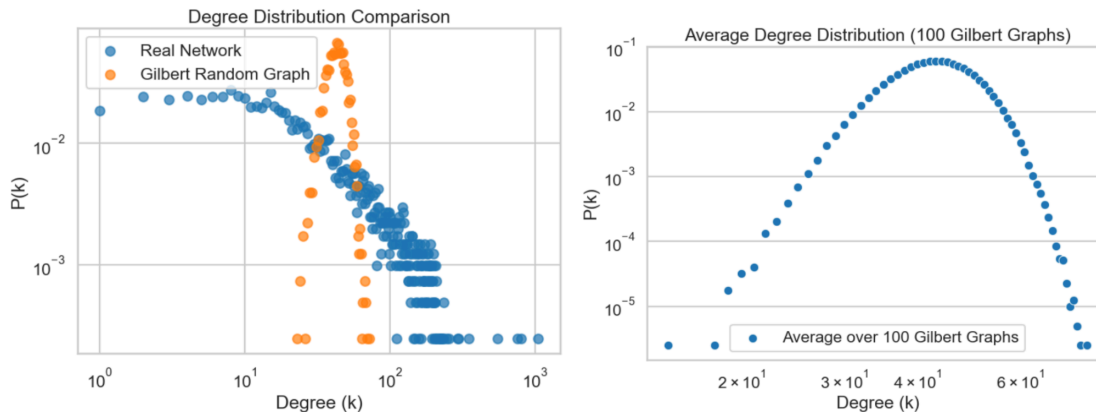
- The weighted degree distribution is highly skewed: most nodes have low weighted degrees, while a few have very high values, including some outliers.
- The log-scale histogram shows nodes concentrated at low degrees, decreasing as degree increases.
- Clustering coefficient declines as weighted degree rises: low-degree nodes tend to have higher clustering, while high-degree nodes show lower values.
- Clustering spans several orders of magnitude, particularly among low-degree nodes.
- **Observations:**
 - The degree distribution is heavy-tailed: a small number of nodes have very high weighted degrees, while most are lightly connected.
 - Negative weighted degrees occur because the dataset is signed, so negative ratings reduce the total degree.
 - Clustering generally decreases as weighted degree increases: low-degree nodes tend to form tightly connected groups, whereas high-degree nodes have lower clustering.
 - The log–log plot highlights the network’s structural heterogeneity.
- **Interpretation**
 - The heavy-tailed weighted degree distribution reflects structural heterogeneity, with a few highly central users.
 - The negative relationship between clustering and weighted degree indicates a hierarchical structure: smaller nodes form tight communities, while highly connected nodes bridge groups.
 - Negative weighted degrees result from signed trust ratings, adding complexity beyond unsigned networks.
 - Overall, the network exhibits key characteristics of real-world weighted social systems.

Question 4

- **Methodology:**

- Parsed the Facebook edge list to compute: N : number of nodes; L : number of edges
- Computed connection probability: $p = (2L) / (N(N - 1))$. Generated a Gilbert random graph $G(N, p)$.
- Computed degree distribution $P(k)$ for: Real network, One random graph, Average over 100 random graphs and plotted all degree distributions on a log-log scale.

- **Results:** Number of nodes (N): 4039; Number of edges (L): 88234; Probability (p): 0.0108



- Real network shows a heavy-tailed degree distribution.
- A single Gilbert graph shows a peaked distribution around the mean degree.
- The averaged distribution over 100 graphs is smooth and resembles a binomial (approximately Poisson) distribution.

- **Observations**

- Real network contains high-degree hubs.
- Random graphs show degrees concentrated near the average.
- No extreme hubs appear in the Gilbert model.
- Averaging over 100 instances reduces fluctuations and stabilises the distribution.

- **Interpretation:**

- The real network is heterogeneous and structurally organised.
- The Gilbert random graph is homogeneous because edges are formed independently.
- Despite matching NNN and the expected number of edges, the random model fails to reproduce the heavy-tailed structure of the real social network.

Question 5

- **Methods:** The Karate Club network was modelled as an undirected graph using an edge list to build an adjacency list. Breadth First Search (BFS) was performed from node 0 using a queue. During traversal, distances, parent nodes, and BFS layers were recorded.
- **Results:** BFS visited all reachable nodes and computed the shortest path distances from the source. The parent mapping formed a BFS tree, with nodes grouped by distance into layers.
- **Observations & Interpretation:** Most nodes were reached within a few layers, indicating strong connectivity and short path lengths. The BFS tree shows shortest paths in the unweighted graph and highlights how information can quickly spread through the network.